

CE/CZ4055: CPS Security

Tutorial 4 - Identifying Attack Surfaces

- (1) A Smart Grid (SG) infrastructure is a well known Cyber Physical system based on digital technology that is used to supply electricity to consumers via two-way digital communication. This system allows for monitoring, analysis, control and communication within the supply chain to help improve efficiency, reduce energy consumption and cost, and maximize the transparency and reliability of the energy supply chain. The smart grid was introduced with the aim of overcoming the weaknesses of conventional electrical grids by using smart net meters. Figure 1 shows a typical scenario in smart grids.

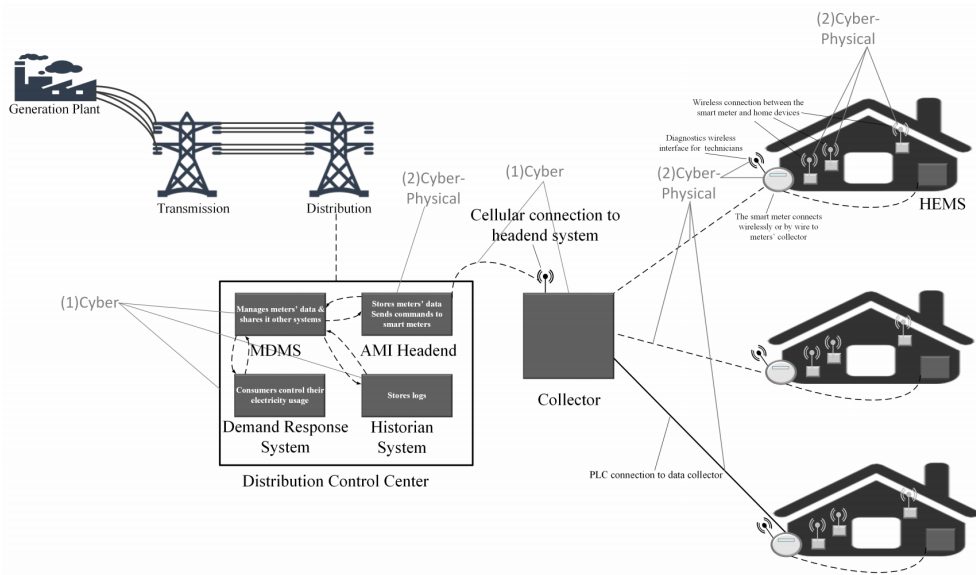


Fig. 1. Smart Grid Architecture

A smart meter is attached to every house to provide utility companies with more accurate electricity consumption data and customers with convenient way to track their usage information. It interfaces with the house's appliances and an energy management system and gives precise information about electricity usage, fluctuating tariff rates etc. It relies on short range wireless communication to interact with the various energy consuming devices. It is connected to a collector, which aggregates information from all the smart meters from a neighbourhood and relays the information to a control center which gets information from many such collectors in the city. All the information that is aggregated is analysed by the company to take suitable actions with respect to the operations in the grid.

- (a) State some vulnerabilities that you can identify at the customer's end. What can happen when one or many of the smart meters are compromised?
- (b) Smart meters are very useful for fine-grained control of usage of electrical appliances at home. Instead of the present or legacy system of only having two rates - a cheap rate for

night and a standard rate for the day, the retail electricity prices can match that of the whole sale electricity prices at an hourly/2-hourly basis. The smart meter can get information about the tariff rates about the present and future (for say the next few days) and if this information is used to control the Home Energy Management System (HEMS) to turn energy demanding appliances only during low tariff rates, it is a win-win situation for both the customers, who save on their bills and also the energy suppliers whose peak energy demand reduces. Though this is a win-win situation for the parties involved, can you find a vulnerability in this scenario?

- (2) Remote keyless Entry is present in all the present day automotive systems (cars, trucks, buses etc.). It works on short range radio transmission usually within a range of 5-20 meters. But, these systems have been shown to be easily breakable using cheap devices. Please watch the video in this link <https://www.youtube.com/watch?v=bXfp8F4J2eI> and explain how such an attack was carried out. What are the possible solutions to the problem?